"To Lane or Not to Lane?"—Comparing On-Road Experiences in Developing and Developed Countries Using a New Simulator *RoadBird*

Md. Masum Mushfiq, Tarik Reza Toha, Saiful Islam Salim, Aaiyeesha Mostak, Masfiqur Rahaman, Najla Abdulrahman Al-Nabhan, Arif Mohaimin Sadri, and A. B. M. Alim Al Islam

***Abstract*— Even though the traffic systems in developed countries have been analyzed rigorously and operated efficiently, the same does not generally hold for developing countries due to inadequate planning, design, and operations of their transportation systems. Because of inherent differences between internal infrastructures, the strategies deployed in developed countries may not be amenable to developing ones. Besides, developing countries' traffic systems are not well-studied in the literature to the best of our knowledge. For example, it is yet to explore how a developed country's lane-based traffic flow would perform in the context of a developing country, which generally experiences non-lane-based traffic. As such, by using our newly developed traffic simulator 'RoadBird,' we investigate outcomes of both lane-based and non-lane-based traffic from the contexts of both developing and developed countries. To do so, we run simulations over real road topologies (extracted from the GIS maps of major cities such as Dhaka, Miami, and Riyadh), considering different scenarios such as lane-based or non-lane-based flows, homogeneous or heterogeneous traffic, with or without pedestrians, etc. We also incorporate various car-following and lane-changing models to mimic traffic behaviors and investigate their performances. While the lane changing dilemma remains an open research question, our experimental evidence indicates: 1) lane-based approaches will not necessarily perform better in the case of currently-adopted non-lane-based scenarios, and 2) non-lane-based strategies may benefit system performance in lane-based scenarios while having heavy mixed traffic. Nonetheless, we reveal several new insights for on-road experiences both in developing and developed countries.**

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Md. Masum Mushfiq, Aaiyeesha Mostak, and A. B. M. Alim Al Islam are with the Next-Generation Computing (NeC) Research Group, Department of CSE, Bangladesh University of Engineering and Technology, Dhaka 1211, Bangladesh (e-mail: masummushfiq@cse.buet.ac.bd; 0419052029@grad.cse.buet.ac.bd; alim_razi@cse.buet.ac.bd).

Tarik Reza Toha is with the Department of Computer Science, University of North Carolina at Chapel Hill, Chapel Hill, NC 27599 USA (e-mail: ttoha12@cs.unc.edu).

Saiful Islam Salim is with the Department of Computer Science, University of Arizona, Tucson, AZ 85721 USA (e-mail: saifulislam@arizona.edu).

Masfiqur Rahaman is with the Halıcıoğlu Data Science Institute, University of California San Diego, La Jolla, CA 92093 USA (e-mail: marahaman@ucsd.edu).

Najla Abdulrahman Al-Nabhan is with the Department of CS, King Saud University, Riyadh 11421, Saudi Arabia (e-mail: nalnabhan@ksu.edu.sa).

Arif Mohaimin Sadri is with the School of Civil Engineering and Environmental Sciences, The University of Oklahoma, Norman, OK 73019 USA (e-mail: sadri@ou.edu).

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***Index Terms*— Computer simulation, road traffic, vehicle routing.**

I. Introduction and Motivation

DHAKA, the capital of Bangladesh, is one of the mega-cities of the world. However, the chaotic traffic system of Dhaka is a major headache for its city dwellers. Traffic congestion in Dhaka wastes around 3.2 million working hours daily, costing the economy billions of dollars [1]. In this condition, saving even a single minute on average can save millions of dollars. Traffic simulation software is a valuable tool for system-wide traffic impact assessment and sustainable policy-making that can come into play in cases such as Dhaka. This can be done by testing the impact of a proposed policy on the intended traffic network. However, the effectiveness and applicability greatly vary depending on how accurately the simulator mimics the desired traffic stream. As Dhaka is a city of unstructured (non-lane based) heterogeneous (high mix of slow and faster-moving vehicles) traffic, its traffic stream experiences diversified on-road scenarios such as pedestrian on the road, illegal parking, deliberate rule violation, jaywalking, and so forth. These are quite different from the structured (lane-based) traffic systems of developed countries with homogeneous traffic in terms of speed. Existing traffic simulators lack the ability to accommodate such on-road scenarios. Hence, the need for an advanced simulator with more customized features comes into play for simulating both structured and unstructured traffic behaviors.

As already pointed out, vehicles not following lanes are among the significant attributes of Dhaka city's traffic. On the contrary, the traffic stream is primarily lane-based in the major cities of developed countries. Therefore, it is quite natural for the following question to arise: "What would happen if Dhaka's traffic system is converted into a lane-based system?" Likewise, what would happen if the major cities in developed countries adapt to non-lane-based strategies? In this study, we investigate this dilemma between "to lane or not to lane" to decide on better system performance. To the best of our knowledge, such a study is yet to be done in the literature, and we are the first to do so.

In our study, we developed a new microscopic traffic simulator named RoadBird, based on its earlier version, formerly DhakaSim [2]. In our current work, we extend the simulator to simulate both lane and non-lane-based traffic. We also incorporate different traffic behavioral phenomena, such as car-following and lane-changing models, to mimic realistic traffic behavior. We investigate their performance and choose the best applicable model for further experimentation. RoadBird can also simulate the actions of other traffic entities such as pedestrians, slow vehicles (i.e., rickshaws), motorbikes, etc. In this study, we run simulations on topologies extracted from Dhaka, Miami, and Riyadh's GIS maps to analyze their performances and trajectories. After that, we use different performance metrics to measure performances in various scenarios.

Based on our work, our contributions are as follows:

- We present a new traffic simulator, RoadBird, capable of simulating both lane and non-lane-based traffic in the presence of both homogeneous and heterogeneous traffic streams. RoadBird can also consider diversified scenarios experienced on roads in developing countries, such as pedestrians on the street and the movement of slow vehicles such as rickshaws.
- We simulate RoadBird to compare the performances in both developed and developing countries by varying traffic load on the network from low to high vehicle generation rate and by varying the ratio of different vehicle types, i.e., the vehicular mix. Here, we use the road networks extracted from Miami and Riyadh as representatives of the developed country's road network and the road network extracted from Dhaka as a representative of the developing country's road network.
- Our observations further analyze and show that non-lane-based systems exhibit better performance by utilizing the road space more efficiently with heterogeneous traffic and narrow roads. However, lane-based systems exhibit better vehicle speed and flow rate with homogeneous traffic and wide streets.
- While lane-based approaches will not necessarily yield more efficient outcomes in non-lane-based scenarios, the other side of the coin is not this impotent. This happens as non-lane-based strategies may benefit system performance in lane-based scenarios with heavy mixed traffic, which we confirm by our analysis.

II. Background and Related Work

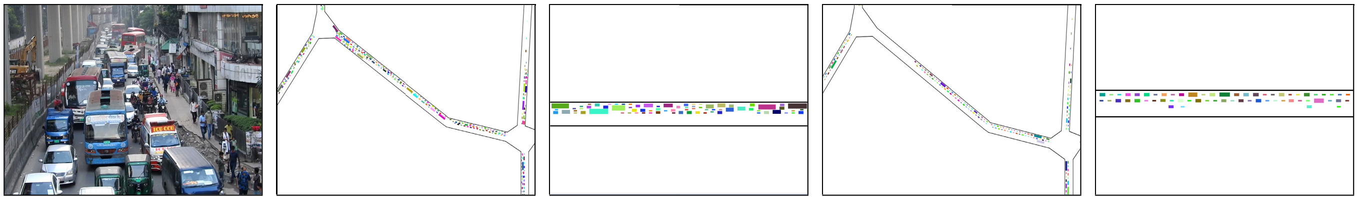
Life in Dhaka is difficult due to its chaotic traffic conditions. Microscopic traffic simulation is widely used and one of the most effective ways to predict traffic behavior. These tools can aid in making important transportation engineering decisions by simulating the proposed decisions and analyzing their impact on the existing traffic network. However, its effectiveness greatly depends on the accuracy of mimicking the intended traffic pattern. Although several microscopic traffic simulators exist in the literature, they fail to mimic the non-lane-based heterogeneous, i.e., unstructured traffic stream of Dhaka city. Hence, we need a customized traffic simulator to simulate the impact of a transportation engineering policy on an unstructured traffic stream. In our previous work, we developed a non-lane-based traffic simulator named DhakaSim [2] to simulate the diversified traffic behavior. However, it lacks the capability of simulating a lane-based structured traffic stream. Moreover, no comparative study has been done on lane- and non-lane-based heterogeneous traffic performance. Hence, we extend the existing DhakaSim traffic simulator to RoadBird to simulate both lane and non-lane-based traffic streams and conduct an extensive experimental study to derive a conclusive remark about "To lane or not to lane". Next, we present some existing work on heterogeneous traffic simulations and related models.

Vedagiri [3] propose a simulator named HETEROSIM to simulate heterogeneous traffic flow considering Indian road traffic. They estimate the saturation flow rate of heterogeneous traffic considering the effect of road. From the simulation result, they found a linear relationship between saturation flow and road width. Arsasan and Dhivya [4] measure one of the fundamental characteristics of traffic flow, i.e., concentration, using HETEROSIM. They argue that the traditional concept of concentration cannot be applied to heterogeneous and non-lane-based traffic and propose a new concept named area occupancy to measure traffic concentration for heterogeneous and non-lane-based traffic.

There are few studies about modeling heterogeneous traffic flow. Arasan and Dhivya [4] propose a simulation framework for the traffic-flow model where the absence of lane discipline in mixed traffic flow conditions is considered. They also describe common issues related to traffic simulation in the context of heterogeneous traffic conditions. Olstam and Tapani [5] discuss a different car-following model only for lane-based traffic. Jin et al. [6] propose a non-lane-based full velocity difference car-following model. They incorporate the lane width effect in car-following models and show that the lateral separation effect greatly enhances the realism of non-lane-based car-following models. Muniruzzaman et al. [7] propose a method to calibrate and validate non-lane-based microscopic simulation models. However, they do not compare lane and non-lane-based traffic systems.

Mathew et al. [8] propose a space discretization-based simulation framework named SiMTraM to address the driver behavioral models in the heterogeneous traffic stream. This simulator can simulate both lane and non-lane-based traffic. However, the proposed simulator has no implementation for the random movement of pedestrians and non-motorized traffic on the road. Agarwal et al. [9] propose an agent-based framework that uses a queuing model to simulate the mobility of only motorized vehicles. Mohan and Ramadurai [10] propose a parsimonious model of heterogeneous traffic that can capture the unique phenomena of the gap-filling behavior of vehicles. They have calibrated and validated the model using field data from an arterial road in Chennai city. Chand et al. [11] develop a dynamic PCU for the candidate signalized intersections catering to mixed traffic conditions in Indian cities. However, the proposed simulator does not deal with the random movement of pedestrians and non-motorized traffic on the road.

All of the studies mentioned above explore different aspects of lane-based structured traffic. However, a comparative study between lane and non-lane-based traffic with heterogeneous motorized and non-motorized traffic streams is yet to be done. Hence, we perform a comparative study among structured and unstructured traffic through our custom-designed and



(a) Real-traffic of Dhaka city (b) Non-lane-based simulation in RoadBird (c) Close-up of non-lane-based simulation in RoadBird (d) Lane-based simulation in RoadBird (e) Close-up of lane-based simulation in RoadBird

Fig. 1. Snapshots of real traffic in Dhaka and simulated traffic in RoadBird.

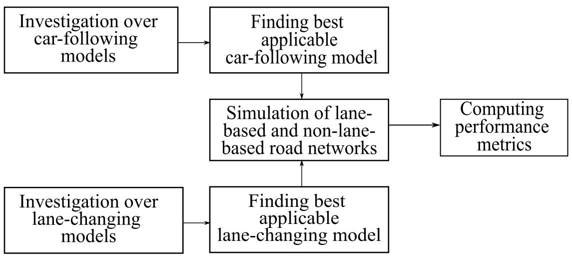


Fig. 2. Flowchart of our comparative study.

developed simulation software named RoadBird in this work. Next, we present the overview of our study.

III. Overview of Our Study

In our study, first, we implement some prominent car-following and lane-changing models from existing literature to simulate both lane and non-lane-based scenarios. We also customize some of the models to better fit the heterogeneous nature of the traffic flow. We investigate these models through simulation and choose the best-fitting model. Then, using the selected models, we compare the performance of lane and non-lane-based road networks. We present the flowchart of our comparative study in Figure 2.

A. Both Lane and Non-Lane Based Traffic Simulation

To gracefully model both lane and non-lane-based traffic, we design a link as a composition of several strips. We can vary the width of the strips through our simulator; through this variation, we achieve the provision of lane and non-lane-based simulations.

The number of strips on a link is calculated as follows,

$$\#\ of\ strips = \left\lfloor \frac{link\_width}{strip\_width} \right\rfloor$$

Each vehicle occupies some strips according to its width, computed as follows,

$$\#\ of\ occupied\ strips = \left\lceil \frac{vehicle\_width}{strip\_width} \right\rceil$$

Our simulator is designed to model both lane and non-lane-based traffic by using strips of varying widths. If a strip is wider than a vehicle, the vehicle will fit entirely within the strip, which we can consider as a lane. Conversely, if a strip is narrower than a vehicle, the vehicle will occupy multiple strips. We create a lane-based scenario by choosing a strip width wider than the maximum vehicle width. In contrast, a non-lane-based scenario occurs when the strip width is narrower than the minimum vehicle width, causing each vehicle to occupy multiple strips. Since the width of vehicles varies by type, the number of strips occupied by each type of vehicle will also vary. Therefore, by varying the strip width, our simulator can simulate both lane and non-lane-based traffic.

B. Car-Following Models

This section will explain the car-following models we chose to implement and their adaptation in non-lane-based scenarios. Car following models in lane-based systems are well studied in the literature. We have adopted a similar methodology to implement the car following models in the lane-based scenario. In our implementation of non-lane-based scenarios, vehicles occupying multiple strips can observe more than one vehicle downstream and upstream on the same set of strips. To determine the leader, we consider the vehicle with the smallest space gap from the current vehicle and perform the necessary calculations accordingly. Additionally, when a vehicle changes strips, the leader and follower measures are updated accordingly.

We implement multiple prominent car-following models and compare the performance by varying these models. Next, we describe each of them.

1) Newtonian Car-Following Model With Explicit Braking: We develop and implement this novel model as a starting point for implementing car-following models. In this model, the vehicles move using Newton's law of motion. We use the following equations to compute the speed and distance of a vehicle.

$$\Delta x = x_{n-1}(t) - s_{n-1} - x_n(t) \tag{1}$$

$$v_n(t+\Delta t) = \begin{cases} v_n + a_n \Delta t, & \text{accelerating} \\ \dfrac{\Delta x}{\Delta t}, & \text{braking} \end{cases} \tag{2}$$

$$x_n(t+\Delta t) = x_n(t) + v_n(t)\Delta t \tag{3}$$

where Δx is the safe distance between leader and subject vehicle, Δt is the time step that is $1s$ for our simulation, s_{n-1} is the effective length of the subject vehicle, and $x_{n-1}(t)$ and $x_n(t)$ are the distance of the leading vehicle $n-1$ and subject vehicle n in a link.

2) Modified Newtonian Car-Following Model: We improve the basic Newtonian Car-Following Model with Explicit Braking as the deceleration of very high magnitude can lead to unrealistic abrupt braking. We update the model to handle acceleration and deceleration uniformly with a single equation, which allows for smooth deceleration when required.

Moreover, we make it more suitable for variable time steps by including the $(1/2)a_n(t)\Delta t^2$ term when calculating the new distance. In addition to that, we limit the maximum deceleration capabilities of vehicles to 8.5 ms^2 according to the study of Kudarauskas [12]. To calculate the speed and distance of a vehicle, we use the following equation, which uses similar notations as the previous model:

$$\Delta x = x_{n-1}(t) - s_{n-1} - x_n(t) \tag{4}$$

$$v_n(t + \Delta t) = v_n + a_n(t)\Delta t \tag{5}$$

$$x_n(t + \Delta t) = x_n(t) + v_n(t)\Delta t + \frac{1}{2}a_n(t)\Delta t^2 \tag{6}$$

3) Gipp's Car-Following Model: Gipps [13] calculates a safe speed with respect to the leader. In this model, speed is computed such that the subject vehicle can be safely stopped if the leading vehicle stops suddenly.

4) Generalized Force Car-Following Model: The Generalized Force (GF) model is a prominent and well-cited car-following model proposed by Helbing and Tilch [14]. We implement this model to observe its behavior in both lane and non-lane-based scenarios. In the GF model, vehicles do not collide with each other and follow the leaders perfectly, but we observe longer trip times because of undergoing stop-and-go waves [15].

5) k-Leader Fuel-Efficient Traffic Model (kFTM): We choose to implement the k-Leader Fuel-Efficient Traffic Model (kFTM) proposed by Awal et al. [16], as it is a recent and robust model with physical interpretations for all its parameters. This model is particularly well-suited for simulating autonomous vehicles, thus enhancing the capabilities of our simulator.

6) Hybrid Model: The Hybrid model amalgamates Newtonian and Gipps' models. Abrupt braking is applied to the Newtonian model, disregarding maximum braking restriction. In cities like Dhaka, where non-lane-based heterogeneous traffic predominates, situations often arise where sharp braking is applied. We introduced this model to capture this driving behavior by integrating the Newtonian and Gipps' models, where vehicles usually move with speed from Gipps' model. Threshold distance refers to the minimum safe distance between vehicles when traffic becomes congested or reaches a state of jam. We measure the likelihood of a collision by the violation of a threshold distance of 2 meters [15], [16]. Such situations arise due to abrupt lane-changing. In these scenarios, braking is applied using Equation 2. We adopt this model for our simulation as validated in Section V.

C. Discretionary Lane Changing Models

A vehicle has to experience three steps to perform a discretionary lane change (DLC)- i) desire to change the current lane, ii) ensure changing lane is feasible, and iii) decision to change lane based on gap acceptance. A vehicle wishes to change its current lane when it cannot move forward or a slower vehicle is in proximity. After a vehicle wishes to change lanes, it calculates the gaps in its adjacent lanes, first the higher speed lane, then the lower speed lane, to determine the feasibility of changing lanes. This feasibility analysis is done by the DLC models implemented in our simulator. Finally, if lane changing is feasible, i.e., there is an acceptable gap in the desired lane, the decision of whether to change lanes or not is calculated. Gap acceptance probability is calculated using Equation 7.

$$p(t) = \begin{cases} 1 - e^{-\lambda(t-\tau)} & \text{if } t > \tau \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

where λ is a co-efficient, τ is the critical time gap, t is the actual time gap which can be computed as, and $t = (g/v_n)$. Here, g is the lead/lag gap, and v_n is the speed of the subject vehicle. The probability that we accept the gap is the product of the probability of both the lead and lag gap are accepted, that is,

$$p(t^{lead}, t^{lag}) = p(t^{lead}) \times p(t^{lag}) \tag{8}$$

We adopt a novel approach to simplifying the conversion of non-lane-based traffic simulation based on lane-based models. As we use variable strip width to convert lane-based traffic to non-lane-based traffic, the existing lane-based models can be adopted in our non-lane-based scenario. In the lane-based scenario, vehicles can change one lane at a time for lateral movements. In non-lane-based scenarios, vehicles can change strips. Vehicles can change multiple consecutive strips if all the strips satisfy the above-mentioned three criteria to seek the opportunity to overtake a slower vehicle or completely change lanes. However, the feasibility and gap acceptance calculations are performed for each strip separately.

We implement two DLC models in our simulator. The reasons that trigger a vehicle to seek lane-changing opportunity are the same for all the models- when it cannot move forward in its current lane or has a slower vehicle in its proximity. We explain the feasibility computation and gap acceptance steps for our implemented models in the following subsections.

1) Straightforward Model: In cities with heterogeneous traffic, it is observed that vehicles behave more abruptly while following leader vehicles and changing lanes. We try to address this behavior in this model. The feasibility of changing lanes is computed using the gaps in adjacent lanes. The gaps between the subject vehicle and the leader and follower vehicle in the target lane are calculated. Lane-changing is feasible if these gaps exceed the accepted threshold distance between two vehicles. The gap acceptance probability for this model is always 1, which means if the target lane has enough gap to accommodate the vehicle, it always shifts to the target lane.

2) GHR Model: In the GHR model, acceleration is used to compute the feasibility of changing lanes and to make decisions. This acceleration is computed using GHR [17] equation as follows.

$$a_n(t) = cv_n^m(t)\frac{\Delta v(t - T_r)}{\Delta x^l(t - T_r)} \tag{9}$$

where a_n is the acceleration of Vehicle n implemented at time t by a driver and is proportional to v_n, the speed of the n_{th} vehicle. Δv and Δx are the speed and space spacing between the leader and subject vehicle, assessed at an earlier time $t - T_r$, where T_r is the driver reaction time. Moreover, c is the sensitivity co-efficient, m is the speed exponent (-2 to $+2$),

TABLE I
VEHICLE GENERATION DISTRIBUTION I

Speed category	Vehicle type	Vehicular modal share (%)	Vehicle distribution (%) Heterogeneous (Dhaka)	Vehicle distribution (%) Homogeneous (Miami and Riyadh)
Slow	Bicycle	9%	55	9
	Rickshaw	89%		
	Van/Cart	2%		
Medium	CNG	83%	40	75
	Bus (2 types)	15%		
	Truck (2 types)	2%		
Fast	Motorbike	88%	5	16
	Car (3 types)	12%		

TABLE II
VEHICLE GENERATION DISTRIBUTION II

Vehicle type	% Vehicle	Vehicle type	% Vehicle
NMV	26	CNG	23
Motorbike	30	Car (3 types)	18
Bus	2		

and l is the distance headway exponent (+4 to −1). We use $c = 15$, $m = 1$, and $l = 2$ for our simulation.

In this model, if the computed acceleration a_n is less than zero (0), the subject vehicle wishes to change the lane. Then, braking is computed using Equation 9. Then the feasibility of lane changing and gap acceptance probability calculation are done using Equation 7 and 8.

D. Vehicle Generation Model

The vehicles are generated according to the negative exponential distributions of vehicular headways. The probability density function is computed by $f(x) = \lambda e^{-\lambda x}$ and the expression for exponential variate headway X can be derived as $X = \mu(-\ln R)$, where μ is the mean headway, R is a random number between 0 and 1.

E. Vehicle Distribution

In this study, we conduct experiments using two types of vehicle distribution- i) Heterogeneous traffic, and ii) Homogeneous traffic distribution. Heterogeneous traffic is the vehicular mix of motorized and non-motorized human or animal-powered slower vehicles. The non-motorized vehicles (NMV) constitute a significant portion of the total traffic volume in heterogeneous traffic distribution. On the other hand, homogeneous traffic is the vehicular mix of different types of motorized vehicles. The percentage of NMVs are insignificant in the case of homogeneous traffic. Examples of NMVs are rickshaws, vans, bicycles, and carts, whereas motorbikes, cars, buses, trucks, and CNGs are examples of motorized vehicles. We use two different vehicular distributions for Dhaka as presented in Table I and Table II. As consolidated traffic distribution for heterogeneous traffic in Dhaka city was unavailable, we collected the modal share and distribution of the different vehicles of Table I from different websites such as rickshaw from Banglapedia (National Encyclopedia of Bangladesh) [18], car and motorbike from the official website of Dhaka School of Economics (DScE) [19], official web-portal of BRTA (Bangladesh Road Transport Authority), etc. Recently, Sabbir [20] have presented a vehicular distribution in some selected intersections of Dhaka city with traffic volume. We show this distribution in Table II.

F. Performance Metrics

To compute the overall performance of a road network under different parameters, we use four performance metrics- average speed on a link, average waiting time on a link, the average vehicle flow rate of a link, and average speed of a vehicle [21]. Now, we describe each of the performance metrics.

1) Average Speed on A Link ($\overline{speed_l}$ Km/Hour): This metric represents the speed with which a vehicle typically crosses the corresponding link and is calculated for all links using Equation 11. Higher speed indicates better performance.

$$\overline{speed_{vl}} = \frac{length_l}{time\_to\_cross_l} \tag{10}$$

$$\overline{speed_l} = \frac{\sum_{v \in S_l} \overline{speed_{vl}}}{|S_l|} \tag{11}$$

where $\overline{speed_{vl}}$ is the average speed of Vehicle v on Link l and S_l is the set of all vehicles that crosses Link l.

2) Average Waiting Time on a Link ($\overline{t_l}$ s): This metric indicates the average time a vehicle has to wait without movement in a link. It is calculated for all links using Equation 12. A lower average waiting time indicates better performance.

$$\overline{t_l} = \frac{\sum_{v=1}^{N} waiting\_time_v}{N} \tag{12}$$

where $waiting\_time_v$ is the individual waiting time of Vehicle v while crossing Link l and N is the total number of vehicles that leave Link l.

3) Average Vehicle Flow Rate of a Link (Vehicles/Hour): This metric indicates the flow rate of a link and is calculated as the average number of vehicles that cross the middle of the corresponding link in an hour. The higher average vehicle flow rate of a link indicates better performance.

4) Average Speed of Vehicles ($\overline{speed_v}$ Km/Hour): This metric represents the speed with which a vehicle travels and is calculated for all vehicles using Equation 14. Higher average vehicle speed indicates better performance.

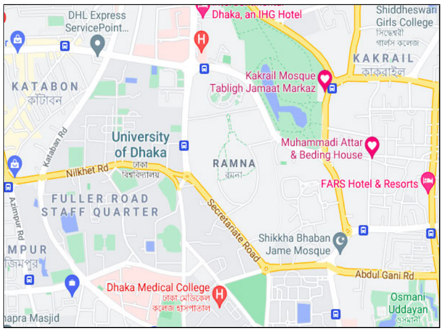
$$\overline{speed_i} = \frac{total\_distance\_traveled_i}{total\_travel\_time_i} \tag{13}$$

$$\overline{speed_v} = \frac{\sum_{v \in S_v} \overline{speed_v}}{|S_v|} \tag{14}$$

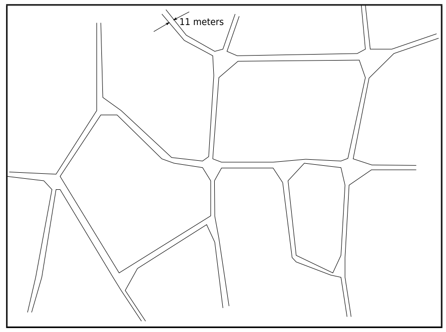
where S_v is the set of all vehicles, $\overline{speed_i}$ is the average speed of the vehicle i, and $|S_v|$ is the total number of vehicles.

IV. EXPERIMENTAL SETUP

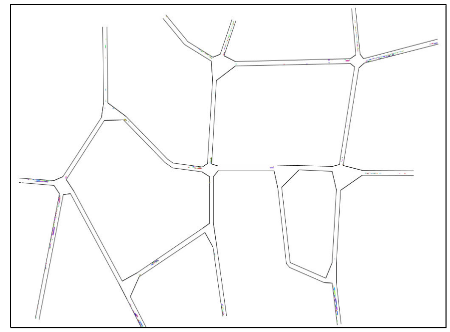
In this section, we describe the experimental setup of our comparative study. In our setup, we vary several significant parameters of our simulator and gather data through simulations for further exploration and performance comparison among lane and non-lane-based road networks. We conduct traffic simulation for 1800s or half an hour for each sample,



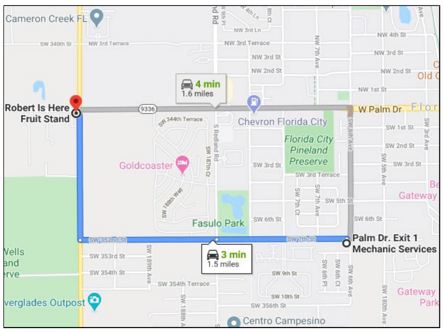
(a) Google map of Dhaka city



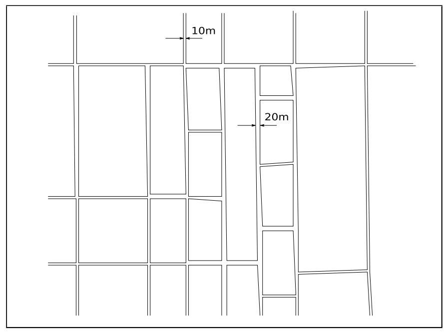
(b) GIS map of Dhaka city



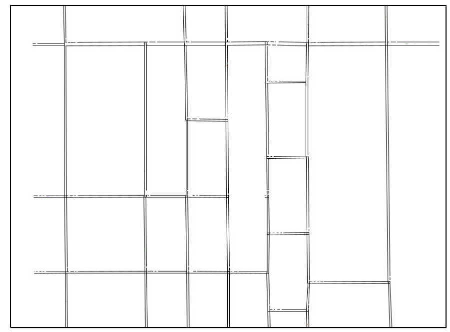
(c) Simulated map of Dhaka city



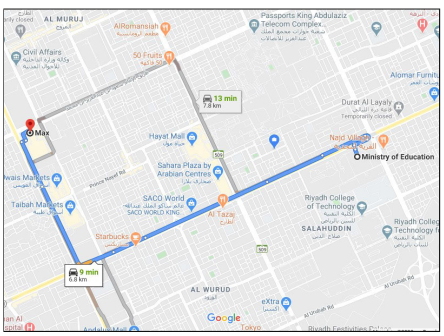
(d) Google map of Miami city



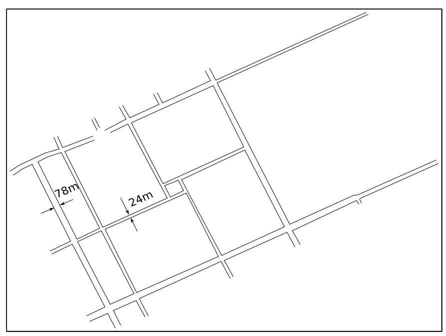
(e) GIS map of Miami city



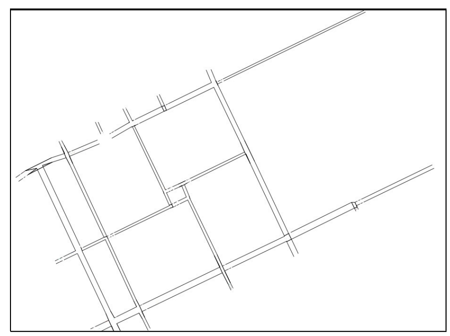
(f) Simulated map of Miami city



(g) Google map of Riyadh city



(h) GIS map of Riyadh city



(i) Simulated map of Riyadh City

Fig. 3. Simulation environment and topologies used in our simulations.

generate ten samples using random number seed 1-10 for each scenario, and take the average of the ten samples for generating graphs and analyzing our result. We can provide different road topologies as input to our simulator. We use three road topologies extracted from the GIS map. They are parts of Dhaka, Miami, and Riyadh city shown in Figure 3. We have three different vehicle generation rates: low, medium, and high. We have three types of vehicles: slow human-powered vehicles (max speed $\leq$ 15 km/hour), vehicles with medium speed (30 km/hour $\leq$ max speed $\leq$ 80 km/hour), and fast vehicles (80 km/hour $\leq$ max speed $\leq$ 120 km/hour). For Dhaka topology, we use 100, 400, and 800 vehicles/hour as low, medium, and high generation rates. On the other hand, we use 500, 1000, and 2000 vehicles/hour as low, medium, and high generation rates respectively for Miami and Riyadh topologies. We use 0.5m and 2.5m as strip widths to simulate non-lane and lane-based road traffic respectively. We enable and disable the irregular crossing of roads by pedestrians through the PedestrianMode parameter. The properties and distribution of vehicles used in the simulations are shown in Table I.

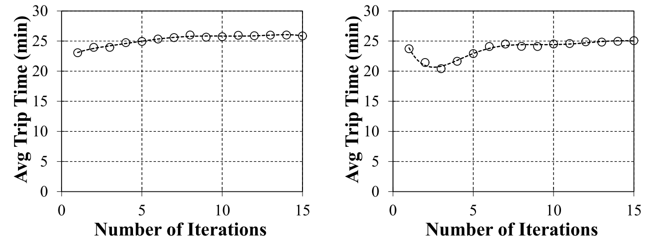
A. Rationale Behind Choosing the Appropriate Car-Following Model

In the case of our simulation for validation of RoadBird, we adopt the Newtonian model with explicit braking, the Modified Newtonian model, Gipp's model, the Generalized Force model, kFTM, and the Hybrid model as the car-following model. Besides, here, we adopt a straightforward model and the GHR model as the lane-changing model. In the case of our simulation for performance comparison after validating RoadBird, we adopt only the GHR model as the lane-changing model.

We have conducted a thorough evaluation of these car-following models to determine the most suitable option for our simulation scenario. Our evaluation shows that the Newtonian model's fixed acceleration and abrupt braking behavior do not accurately represent real-world driving scenarios. Similarly, the Gipps model's lack of robustness in the presence of sensor errors [15] and inability to avoid collisions make it unsuitable for our simulation scenario. While not leading to collisions, the GF model fails to optimize the use of empty roads in low to medium traffic densities (Figure 5

TABLE III
SIMULATION PARAMETERS FOR VALIDATION

Simulation time (minutes)	40	Vehicle generation rate (vehicle/hour)	Low traffic density	100
# of iterations in each case	15		Medium traffic density	400
# of intersections	8		High traffic density	800
# of vehicle generating nodes	8	Pedestrian mode		On
# of links	18	Strip width (m)		0.5
Distance between the two destinations (km)				4.1



(a) Average trip time with number of iterations for motorbike (b) Average trip time with number of iterations for car

Fig. 4. Sensitivity of average trip time with the number of iterations in RoadBird.

and Figure 6). From the study conducted by Awal et al. [15], we conclude that the kFTM model best simulates autonomous vehicles. However, we have chosen the Hybrid model for our simulation scenario involving heterogeneous traffic in Dhaka. This model assumes appropriate acceleration from Gipps' model, does not create any collision, accurately represents real-world driving behavior by optimizing the use of empty roads, and is statistically sound. Thus, after meticulously evaluating and considering all aspects, we conclude that the Hybrid model is the most suitable option for our simulation scenario.

V. VALIDATION OF ROADBIRD

Validation is an integral part of ensuring the credibility of a simulator, as it checks how accurately the simulator represents the real world. Accordingly, we perform validation of the outcomes of RoadBird in comparison to real-world cases.

A. Data Collection for Validation

In our validation process, we choose travel time as a measure of effectiveness (MOE). To measure the travel time, we select a route in Dhaka city from Shankar to Palashi. To get actual travel times on the route chosen over several periods, we collect travel time data in both directions over the route for two types of vehicles, namely cars, and motorbikes. In collecting travel time data, we log real-time travel times from Google Maps, as it provides live travel time. Besides, we have extracted the selected route from the GIS map to run our simulations in RoadBird. We perform several iterations over the simulator and collect outcomes.

We use three different vehicle generation rates in our simulator. Furthermore, we divide the whole duration of a day (24 hours) into three parts based on on-road traffic density based on our day-to-day experience - 1) 10:00 PM - 09:00 AM for low traffic density, 2) 09:00 AM - 04:00 PM for medium traffic density, and 3) 04:00 PM - 10:00 PM for high traffic density. We collect 63 real travel data from Google Maps over these recent durations. We do not use travel time data during the pandemic period when the lockdown period is in operation due to COVID-19, as traffic density is substantially less than usual. Out of the 63 travel data from Google Maps, we collected 34 travel data during the pre-Corona pandemic era; the rest were during the Corona pandemic era. We use some data from the COVID-19 pandemic era, as these apply to low and medium traffic density. However, we only collect high-traffic density data from the pre-COVID-19 pandemic era. We present our simulation parameters in Table III.

Besides, we collect travel times from 45 iterations in total in our simulation, where we consider 15 iterations for each of the three different traffic densities. Though the results converge from around ten iterations onward, as shown in Figure 4, we conduct 15 iterations to present more stable results.

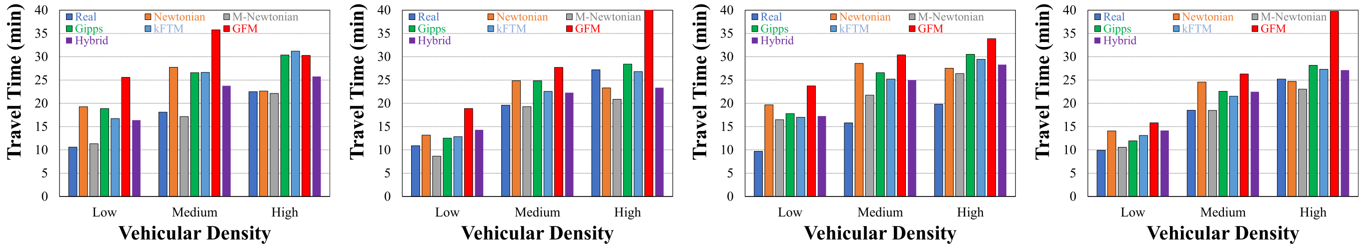
B. Comparison of All Car-Following Models With Real Data

We conduct experiments to compare all the car-following models with real data using both the vehicular distribution from Table I and Table II. We employ the straight-forward lane-changing model to conduct these experiments due to its simplicity, providing a consistent basis for comparison across all car-following models under consideration. The results of the simulation are depicted in Figure 5 and Figure 6.

Upon examining the results of our simulations, we find that both of the Newtonian Models exhibit shorter travel times than the other models. This can be attributed to their constant use of fixed maximum acceleration and deceleration. The Modified Newtonian Model, which takes into account the $(1/2)a_n\Delta t^2$ term in its distance computation results in even shorter travel times than the original Newtonian Model. The GF model consistently exhibits the highest travel times in almost all cases due to its weak acceleration property. The Gipps and kFTM models perform similarly in both vehicle distribution scenarios. Interestingly, the Hybrid model closely mimics the travel time observed in real-world data when using vehicle distribution I from Table I. With an increase in the percentage of cars and motorbikes in vehicle distribution II from Table II, the Hybrid model's travel time slightly increases. However, as validated in Section V-E, this deviation is not statistically significant.

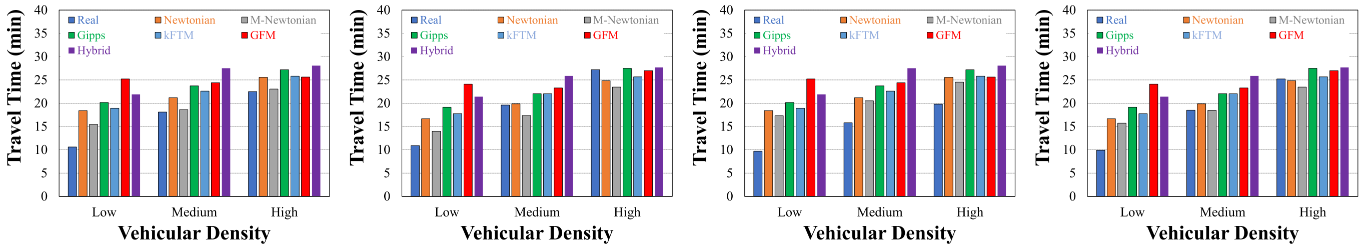
C. Performance Comparison

We present our simulation results for validation in Figure 7. Figure 7a and 7b present average travel times of cars under low, medium, and high traffic density in both directions of the selected route. Besides, Figure 7c and 7d present average travel times of motorbikes under low, medium, and high traffic density in both directions of the selected route. Here we simulate non-lane-based heterogeneous traffic similar to the real traffic of Dhaka using the Hybrid car-following and GHR lane-changing models with vehicle generation distribution I. This is labeled as RoadBird in Figure 7. Results presented in



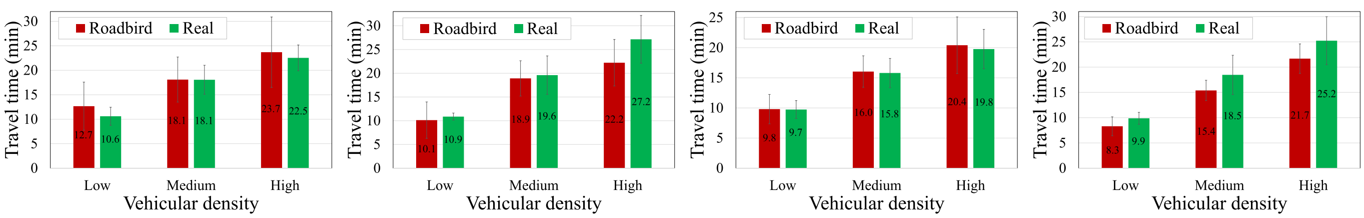
(a) Shankar to Palashi with car (b) Palashi to Shankar with car (c) Shankar to Palashi with motorbike (d) Palashi to Shankar with motorbike

Fig. 5. Travel time comparison among all CF models and observed data for car and motorbike in Shankar to Palashi route - with Vehicle Generation Distribution I.



(a) Shankar to Palashi with car (b) Palashi to Shankar with car (c) Shankar to Palashi with motorbike (d) Palashi to Shankar with motorbike

Fig. 6. Travel time comparison among all CF models and observed data for car and motorbike in Shankar to Palashi route - with Vehicle Generation Distribution II.



(a) Shankar to Palashi with car (b) Palashi to Shankar with car (c) Shankar to Palashi with motorbike (d) Palashi to Shankar with motorbike

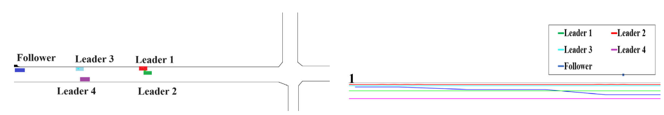
Fig. 7. Travel time comparison between observed and simulated data for car and motorbike in Shankar to Palashi route - with Vehicle Generation Distribution I.

Figure 7 show that travel times obtained through simulations of RoadBird and obtained in real scenarios closely match each other. To further dig into how far they match each other, we perform several statistical analyses of the travel time data, which we discuss in the following sections.

D. Microscopic Validation of Hybrid Model

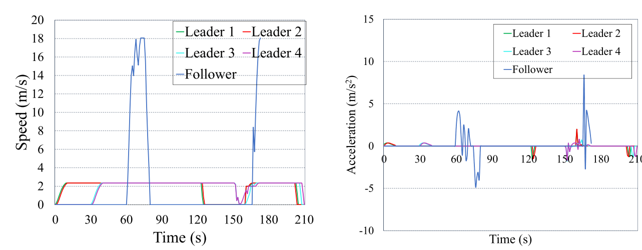
To further validate the microscopic behavior of our simulator, we opt for a four-road intersection scenario. Drawing inspiration from Kanagaraj and Treiber [22], who proposed several idealized cases to validate microscopic car-following models within heterogeneous traffic environments, we experiment with three such idealized scenarios. These are depicted in Figure 8a. In the first scenario, a faster following vehicle navigates past two slower leading vehicles (Leader 3 and Leader 4) by utilizing the gap between them. In the second scenario, the same following vehicle maneuvers around another set of slower leading vehicles (Leader 1 and Leader 2) to the right. Subsequently, in the final scenario, after overtaking these vehicles, the follower comes to a halt at an intersection (200 m from the start) and proceeds across as soon as the traffic light turns green, thereby simulating a bottleneck situation that prompts abrupt speed changes. The trajectories of the vehicles for scenarios 1 and 2 are clearly illustrated in Figure 8b.

Observations from Figure 9a and Figure 9b provide insights into the speed and acceleration patterns of the vehicles



(a) Starting position (b) Vehicle trajectories

Fig. 8. Special scenario for microscopic validation.



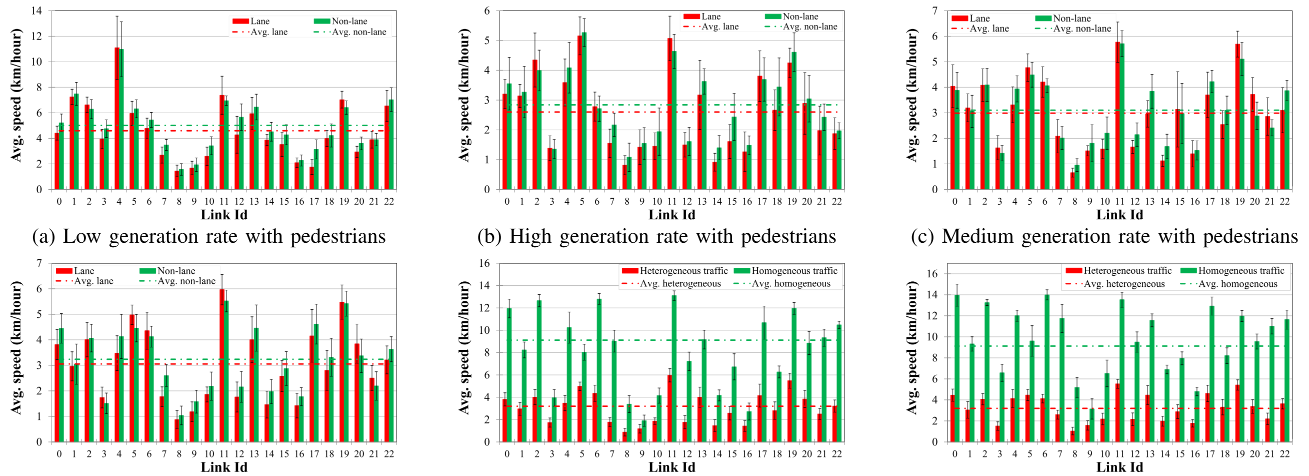
(a) Speed of the vehicles over time (b) Acceleration of the vehicles over time

Fig. 9. Validation of the simulator using microscopic measures.

involved. Notably, the follower reduces its speed while maneuvering a lane change, then accelerates to surpass the slower leaders. Following this overtaking maneuver, it decelerates appropriately upon approaching the intersection, pausing until the signal permits further movement. Once the traffic light turns green, all vehicles involved accelerate, cross the intersection, and make their exit.

TABLE IV
SUMMARY RESULT FOR T-TEST AND ERROR COMPUTATION

Route	Shankar to Palashi						Palashi to Shankar					
Vehicle type	Car			Motorbike			Car			Motorbike		
Vehicle density	Low	Medium	High	Low	Medium	High	Low	Medium	High	Low	Medium	High
p-value of t-test	0.281	0.134	0.69	0.009	0.381	0.802	0.637	0.242	0.12	0.62	0.021	0.018
p-value of K-S test	0.375	0.181	0.375	0.009	0.626	0.925	0.003	0.181	0.181	0.375	0.076	0.009
ME (min)	-1.261	-2.325	-0.712	1.581	-0.894	-0.318	0.447	1.861	2.842	-0.459	2.907	4.403
MAE (min)	2.046	2.359	3.36	1.581	0.898	0.931	1.858	2.217	3.248	1.349	2.907	4.576
RMSE (min)	2.753	3.214	3.78	1.659	1.152	1.337	2.99	2.464	3.875	2.315	3.352	5.078
MAPE (%)	18.333	11.933	15.153	16.64	5.32	5.533	16.84	11.367	11.233	12.467	14.547	17.327
RMSPE (%)	24.199	15.314	17.38	17.611	6.476	8.692	25.72	12.811	12.994	19.78	16.056	18.353



(a) Low generation rate with pedestrians (b) High generation rate with pedestrians (c) Medium generation rate with pedestrians

(d) Medium generation rate without pedestrians (e) Lane-based network with medium generation rate (f) Non-lane-based network with medium generation rate

Fig. 10. Performance comparison based on average speed on a link for Dhaka topology under various combinations.

E. Statistical Analysis Over Real and Simulated Outcomes

For statistical validation, we consider a two-sample t-test and Kolmogorov-Smirnov (K-S) test with a 5% level of significance [23]. The null hypothesis in these tests is that the travel time observed from the natural world and the travel time computed by our simulator come from the same distribution, and the alternative hypothesis is they come from different distributions. Then, we compute the p-value. From the p-value, we can estimate how closely the simulation matches the real world. Goodness-of-fit measures are generally used to calculate the overall performance of simulation models. According to Toledo and Koutsopoulos [24], Ni et al. [25], popular goodness-of-fit measures are Mean Error (ME), Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and Root Mean Square Percentage Error (RMSPE), etc. We can quantify the overall error of our simulator with these statistics. So, we have computed these statistics, and we summarize the results in Table III.

F. Qualitative Outcomes of Our Statistical Analysis

From the p-value of our two-sample t-test and two-sample K-S test, with a 5% level of significance, we can see that 9 out of 12 cases have a p-value more significant than our chosen Level of Service(LOS). This clearly states that our simulation closely matches the real-world traffic scenario. Mean Error (ME) indicates the existence of systematic under- or over-prediction in the simulated measurements. From our calculation, our simulator is not inherently biased in any direction. From Mean Absolute Error (MAE) and Root Mean Square Error (RMSE), our predicted travel time varies from less than 1 minute to at most 5 minutes. From Mean Absolute Percentage Error (MAPE), we can say that the accuracy of our simulator varies from 82-95% with an average accuracy of 88%. From Root Mean Square Percentage Error (RMSPE), we find that the average accuracy of our simulator is 85% with a peak value of 94%. Though RMSPE penalizes the outliers more, we have only two cases where the accuracy drops to 75%.

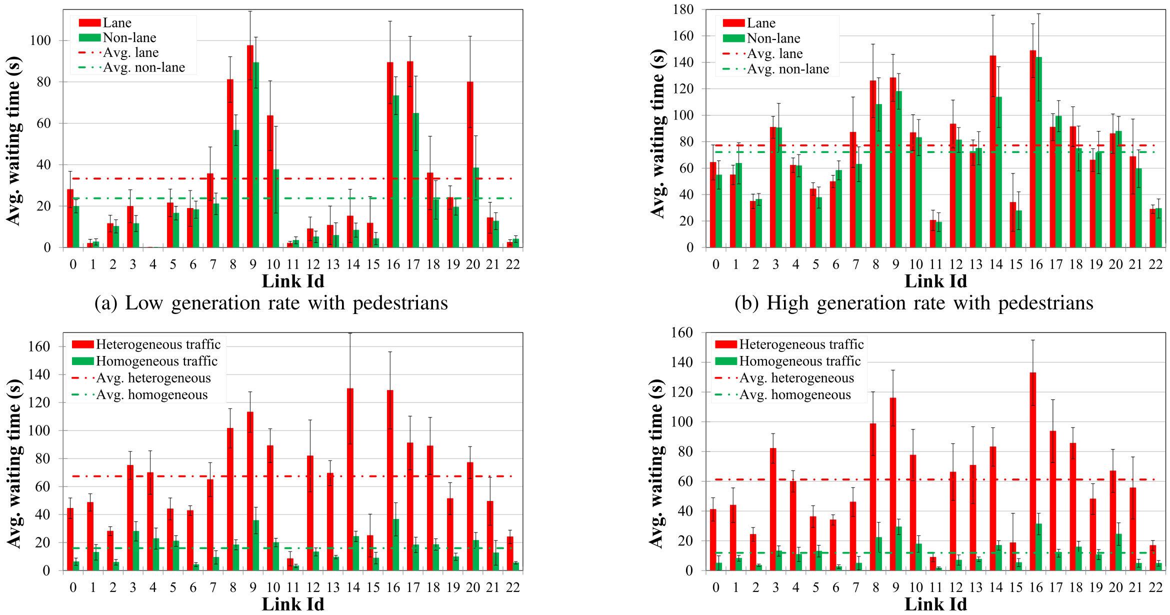
VI. RESULTS ON COMPARATIVE PERFORMANCE STUDY

After validating our simulator, we simulate lane and non-lane-based traffic systems with the parameters described in the experimental setup part. Now, we present their comparative performance concerning different metrics.

A. Performance Comparison for Dhaka Topology

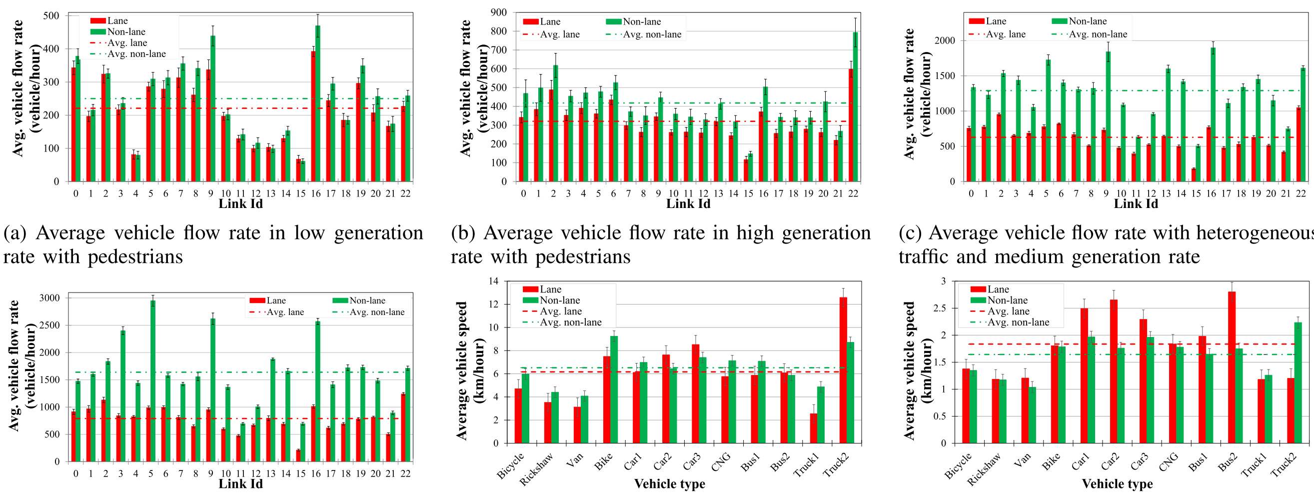
As a representative city of the developing world, we consider Dhaka to simulate both lane and non-lane-based traffic through different parameters of RoadBird. The following subsections discuss the comparative performance of lane and non-lane-based traffic in Dhaka.

1) Variation in The Average Speed on a Link: We present a performance comparison based on average speed on a link for Dhaka topology under various combinations in Figure 10a, 10b, 10c. According to this figure, the average



(a) Low generation rate with pedestrians
(b) High generation rate with pedestrians
(c) Lane-based network with medium generation rate
(d) Non-lane-based network with medium generation rate

Fig. 11. Performance comparison based on average waiting time on a link for Dhaka topology under various combinations.



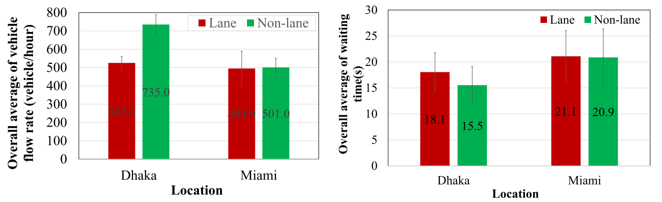
(a) Average vehicle flow rate in low generation rate with pedestrians
(b) Average vehicle flow rate in high generation rate with pedestrians
(c) Average vehicle flow rate with heterogeneous traffic and medium generation rate
(d) Average vehicle flow rate with homogeneous traffic and medium generation rate
(e) Average speed of the vehicles in low generation rate with pedestrians
(f) Average speed of the vehicles in high generation rate with pedestrians

Fig. 12. Performance comparison based on the average vehicle flow rate of a link and average speed of the vehicles for Dhaka topology under various combinations.

speed on a link decreases with an increasing vehicle generation rate. Besides, the average speed on links is below 10 km/hour, which matches the World Bank report [1], i.e., the average speed of Dhaka city is around 7 km/hour. We can also see that irrespective of generation rate, the average speed for the non-lane road network is higher than that of lane-based road networks. As the roads of Dhaka are narrow and most of the vehicles are small such as rickshaws, motorbikes, cars, etc., non-lane-based systems better utilize the road space. Besides, as the traffic is heterogeneous and slow vehicles prevail in high ratios, speed is mainly controlled by the slow vehicles. This is the reason for the higher speed in the non-lane-based traffic stream. Here, we enable the pedestrians and adopt heterogeneous vehicle distribution as shown in Table I. We compare the performance by keeping the generation rate fixed to a medium rate and enabling or disabling pedestrians on the road. In that case, the performance does not change significantly due to heterogeneous traffic, which we see in Figure 10c and 10d. Let us compare the performance by changing the distribution of vehicles into homogeneous ones. Performance significantly improves for lane and non-lane-based traffic as slow vehicles are removed from the traffic stream, which is evident from Figure 10e and 10f.

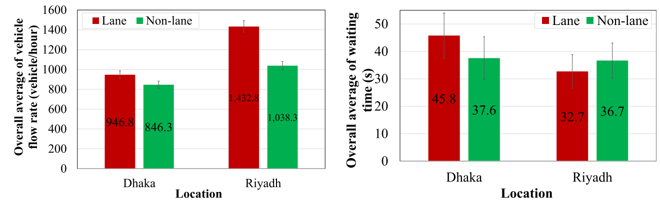
2) Variation in The Average Waiting Time on a Link: We compare performance based on average waiting time on a link for Dhaka topology under various combinations in Figure 11. According to this figure, the average waiting time increases with the expansion in vehicle generation rate, and the average non-lane-based waiting time is less than lane-based waiting time. If we change the distribution into homogeneous traffic, waiting time on links significantly reduces.

3) Variation in The Average Vehicle Flow Rate of A Link: We present a performance comparison based on the average vehicle flow rate of a link in Figure 12a and Figure 12b.



(a) Average vehicle flow rate with medium generation rate (b) Average waiting time with medium generation rate

Fig. 13. Performance comparison between Dhaka and Miami traffic distribution in Miami topology.



(a) Average vehicle flow rate with medium generation rate (b) Average waiting time with medium generation rate

Fig. 14. Performance comparison between Dhaka and Miami traffic distribution in Riyadh topology.

These figures show that the average vehicle flow rate increases with an increasing generation rate. Besides, the vehicle flow rate significantly increases for both lane and non-lane-based traffic networks when traffic distribution switches from heterogeneous to homogeneous, as shown in Figure 12c and Figure 12d.

4) Variation in The Average Speed of Vehicle: We present the comparative performance based on the average speed of vehicles in Figure 12e and Figure 12f. We can see that the average speed of vehicles decreases with the increase in vehicle generation rate. Besides, although non-lane-based networks perform better at a low generation rate, lane-based networks outperform non-lane-based networks at a high generation rate, which supports our practical experience. When the generation rate is high, the structured organization of lane-based traffic helps it to achieve better speed.

B. Performance Comparison for Miami Topology

This subsection presents the comparative study between Dhaka and Miami traffic systems by simulating lane and non-lane-based traffic on Miami topology. Although non-lane-based traffic cannot be seen in Miami, Miami's traffic system becomes lane-less during a hurricane evacuation. Hence, we will try to know what will happen if Dhaka's unstructured traffic runs in Miami. Next, we compare the following performance metrics for lane and non-lane-based traffic in the Miami topology.

1) Variation in The Average Vehicle Flow Rate on A Link: We depict average vehicle flow rates on links in Figure 13a. The figure clearly shows that average vehicle flow rates for the Dhaka distribution are higher than that of the Miami distribution. The gap between the Dhaka traffic distribution and Miami traffic distribution becomes more prominent in the case of the non-lane-based network (Figure 13b). Since most of the vehicles in Dhaka distribution are small and the roads of our chosen Miami topology are narrow as well, lane-less traffic maximizes road space utilization; hence, it moves faster than structured traffic.

2) Variation in Average Waiting Time on A Link: The average waiting time on the link is shown in Figure 13d. From the figure, it is clear that Dhaka's traffic system experiences less waiting time than Miami's one, which is more prominent in the lane-less network as before.

C. Performance Comparison for Riyadh Topology

In this subsection, we compare the performance of lane-based systems with homogeneous traffic and non-lane-based systems with heterogeneous traffic on Riyadh topology. Figure 14a and 14d show the performance comparison between Dhaka's traffic (lane-less) and Riyadh's traffic (lane-based). Unlike Miami, lane-based traffic performs better in Riyadh in vehicle flow rate and waiting time. Since the roads of Riyadh are much broader and more prolonged than Dhaka roads, as shown in Figure 3, it can accommodate a lane-based system better.

Here, the critical aspect, as observed in the simulation, is the fact that the much broader and longer roads of Riyadh permit more vehicles (due to the wider roads having more lanes) to enable high speed for a more extended period of time (due to the longer roads), which was not the case for the narrower and shorter streets of Dhaka and Miami. Besides, the wider roads of Riyadh enable more lateral movements in the case of non-lane-based activity, resulting in a worse performance than that of lane-based movement.

VII. Conclusion and Future Work

The current traffic system in developing countries is less well-disciplined than in developed countries. Hence, a comprehensive comparative study of these systems has received little attention from the academic community. There had been a significant gap in the literature in the contexts mentioned above, and existing studies have yet to address most of the important issues in these scenarios. Our work extensively studies the dilemma of setting up hypothetically lane-based traffic systems in an existing non-lane-based setup. The average speed on links and vehicle flow rate on links decrease with increasing vehicle generation rates for both lane and non-lane-based systems. Waiting time on links increases with increasing vehicle generation rates for both lane and non-lane-based systems. Irregular road crossings by pedestrians do not significantly alter the performance of heterogeneous traffic streams but slightly decrease the performance of homogeneous ones. With heterogeneous traffic, non-lane-based traffic systems perform better when roads are narrow. Lane-based traffic outperforms non-lane-based traffic when the traffic stream is homogeneous, and roads are wide enough to accommodate lane-based systems.

From the above summary, only converting the traffic system of Dhaka into a lane-based system does not guarantee better performance due to the small, slow, human-powered vehicles, and narrow roads of Dhaka city. In addition, non-lane-based scenarios may provide better system performance in many cases, such as emergency evacuations where one may expect a fair combination of slow and faster-moving vehicles. For example, lane reversal or contra-flow often come into practice during major evacuations; however, such scenarios may benefit from adopting non-lane-based strategies. Moreover, RoadBird is capable of modeling behavioral phenomena such as pedestrians, bicycles, and signal control, among others.

While we do not claim that establishing non-lane-based systems in all scenarios will eradicate the current problems overnight, we have clear empirical evidence that wide road networks can indeed benefit from lane-based systems. However, our experiments also show that narrower roads perform worse in lane-based systems. A natural future direction of our work would be to update our model with more complex and realistic parameters (e.g., traffic safety, road intersections, traffic signaling, and pedestrians). Another potential extension could be using a combined setup of both lane and non-lane-based systems in the same topology. Moreover, such simulation techniques could lead to new research directions for identifying more efficient strategies in major crises such as a hurricane or wildfire evacuation where demand typically exceeds capacity.

Acknowledgment

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Md. Masum Mushfiq received the B.Sc. (Engg.) and M.Sc. (Engg.) degrees in computer science and engineering from the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh. He is currently a Lecturer with the Department of Computer Science and Engineering, BUET. His research interests include intelligent transportation systems, modeling and simulation, and computer security.

Tarik Reza Toha received the bachelor's and master's degrees from the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology. He is currently pursuing the Ph.D. degree with the Department of Computer Science, University of North Carolina at Chapel Hill, USA. His research interests include embedded systems, intelligent transportation systems, and computer vision.

Saiful Islam Salim received the B.Sc. and M.Sc. degrees from the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology (BUET). He is currently pursuing the Ph.D. degree with the Department of Computer Science, The University of Arizona, USA. His research interests include intelligent transportation systems, multi-objective optimization, computer security, and natural language processing.

Aaiyeesha Mostak received the B.Sc. degree from the Department of CSE, BUET, where she is currently pursuing the M.Sc. degree in computer science, under the supervision of Prof. A. B. M. Alim Al Islam. Her research interests include neural computing, machine learning, human–computer interaction, and the Internet of Things.

Masfiqur Rahaman received the bachelor's and master's degrees from the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology, under the supervision of Prof. A. B. M. Alim Al Islam. He is currently pursuing the Ph.D. degree in data science with the University of California at San Diego, San Diego. His research interests include intelligent transportation systems, wearable sensing, mobile health, and ubiquitous computing.

Najla Abdulrahman Al-Nabhan was appointed as a Visiting Professor with Nanjing University, China, in 2019. She is currently an Associate Professor with King Saud University (KSU). She has more than 60 published journal articles that are indexed in Web of Science Core Collection (ISI). She also the Senior Consultant of many governmental and private sectors in Saudi Arabia in the field of innovation and research collaboration, including the Ministry of Education, Princes Nora University, and Riyadh Chamber. Her research interests include communication modeling and intelligence, the Internet of Things, unmanned aerial vehicles, and wireless sensors networks. She was awarded U.S. patent in 2013.

Arif Mohaimin Sadri received the Ph.D. degree in civil engineering (transportation), network science, and social science from the Lyles School of Civil Engineering, Purdue University. He is currently an Assistant Professor with the School of Civil Engineering and Environmental Sciences, The University of Oklahoma. His research interests include how transportation systems critically depend on social and other physical systems in the context of natural and man-made hazards. He develops data-driven and network-based solutions to enhance bottom-up resilience in complex, interdependent systems. His research is funded by the National Science Foundation, United States Department of Transportation (DOT), Oklahoma and Florida DOT, and United States Agency for International Development.

A. B. M. Alim Al Islam received the B.Sc. and M.Sc. degrees in computer science and engineering from Bangladesh University of Engineering and Technology, Dhaka, Bangladesh, and the Ph.D. degree in computer science and engineering from the School of ECE, Purdue University, West Lafayette, USA, in 2012. He is currently a Professor with Bangladesh University of Engineering and Technology. His research interests include wireless networks, embedded systems, modeling and simulation, computer networks security, and reliability analysis.